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MNT/013 – Policy for the Management and Control of Vegetation Near Overhead Lines

1 Purpose

The purpose of this document is to set out the company's policy for the management of vegetation near overhead lines. Adherence to this policy will ensure compliance with our company goals and our legal and regulatory obligations for the safety, security and continuity of operation of the distribution network.

This document more specifically describes how we manage the vegetation activity to support our compliance with:

- The Electricity Safety, Quality, and Continuity Regulations 2002 No. 2665 (Issued Jan 2003);
- The Electricity Safety, Quality, and Continuity (Amendment) Regulations 2006 No. 1521 (Issued Jun 2006);
- The Health and Safety at Work Act 1974 (Issued 1974); and
- The Electricity at Work Regulations 1989 no. 635 (Issued Apr 1989).

In addition, the document describes the approach taken for vegetation management in accordance with ENA-ETR 132 – Improving Network Performance Under Abnormal Weather Conditions by Use of a Risk Based Approach to Vegetation Management Near Electric Overhead Lines.

2 Scope

This policy document applies to all overhead line voltages up to and including 132kV.

This document applies to all aspects of the implementation, management and control of the vegetation management process within the distribution business. The document provides an overview of the general rules for defining the approach to the management of vegetation in the event that it encroaches into overhead lines. It goes on to discuss the level of response to vegetation encroachment into overhead lines. It also includes the ongoing maintenance and control of the process defined.

This document is aimed at setting policy and offering sources of information to the whole business in a form that is independent of the organisational structure. It enables all of the different functions within the business to make informed decisions that will ensure compliance with all aspects of the vegetation management activity.

At the time of writing this document it is envisaged that a company policy for vegetation management will always be required by the Distribution business. Hence, the drafting intention is to produce a policy that is capable of standing within the company's policy portfolio for an indefinite period.

3 Policy

3.1 Assessment of Relevant Drivers

As vegetation growth encroaches into overhead line clearances, a potential hazard to safety and quality of supply performance is created. If vegetation encroachment were allowed to continue, there would be an increased likelihood of contact with bare conductors by a member of the public, particularly children by

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using trees as a climbing aid. In addition customer performance would be impacted, as faults would increase due to trees increasingly coming into contact with overhead lines.

It is the vision of CE Electric UK to be within the top 10% of utilities for safety and customer service performance. This requires us to manage vegetation to avoid danger and interruption to supply.

We also have general obligations to keep our staff and third parties safe, so far as is reasonably practicable, under the Health and Safety at Work Act 1974 and the Electricity at Work Regulations 1989.

The Electricity Supply, Quality and Continuity Regulations 2002 include a duty relevant here, specifically:

- *Regulation 18(5): No overhead line shall, so far as is reasonably practicable, come so close to any building, tree or structure as to cause danger.*

This regulation imposes a continuous duty to keep overhead lines clear from trees for the purpose of avoiding danger. The associated guidance notes state that duty holders **may** demonstrate compliance with this requirement by complying with Energy Networks Association's Standard 43-8 Overhead Line Clearances. The guidance also states that the Department for Business, Enterprise and Regulatory Reform (BERR) has an expectation that duty holders would operate tree cutting programmes with sufficient frequency to ensure that trees do not become a source of danger, for example due to children climbing trees near overhead lines.

The 2002 Regulations were amended by the Electricity Supply, Quality and Continuity (Amendment) Regulations 2006 (to create the 2006 Regulations), specifically creating a new requirement for duty holders, specifically:

- *Regulation 20A: A generator or distributor shall, so far as is reasonably practicable, ensure that there is no interference with or interruption of supply caused by an insufficient clearance between any of his overhead lines and a tree or other vegetation.*

This regulation imposes a continuous duty to keep overhead lines clear from trees so as to prevent danger and interruption to supply. The guidance notes to the 2006 Regulations make it clear the date by which compliance is required to be achieved is 31 January 2009 and that ENA-TS 43-8 clearances are expected by BERR to demonstrate compliance. The guidance notes also refer to the application of the risk-based approach documented in ENA-ETR 132.

3.2 Key Policy Requirements

The objectives of the policy are:

- To achieve and maintain appropriate clearances between vegetation and overhead lines;
- To ensure compliance with all relevant legislation and statutory obligations for the safety of third parties and the operation of the distribution network;
- To mitigate the safety and quality of supply risks associated with not having an appropriate vegetation management process in place;
- To ensure a measured and appropriate level of response is applied if vegetation encroachment into overhead lines has been identified; and
- To address customer expectations and requirements with respect to vegetation management.

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In order to run this policy effectively and consistently across the company, there are two key requirements. Firstly, the identification of vegetation encroaching into overhead lines. And secondly, the application of a consistent level of response to vegetation encroachment. These key requirements are described in the following section of the document.

3.2.1 Identification of Vegetation Encroaching into Overhead lines

Network Services shall inspect for (amongst other things) vegetation encroachment in accordance with the Policy for the Inspection and Maintenance of Overhead Lines – MNT/004. Accurate gathering and recording of information shall be carried out as part of this activity, to include next return date, 'High Risk' areas (see section 3.2.2), sensitive customer issues, tree amenity issues etc.

Management performance indicators shall be employed to monitor the level of risk that has been mitigated by application of this activity.

3.2.2 Standard Clearance Requirements

The fundamental requirement for all non-effectively insulated overhead lines is to maintain ENA-TS 43-8 (This document has been incorporated into an internal document, "Guidance on Overhead Line Clearances - NSP/004/011") clearances at all times for all lines. Therefore:

- For overhead lines at 33kV or below:
 - o 0.8 metre clearance shall be maintained between the overhead line conductor and that part of the tree under/adjacent to the line that is unable to support a ladder or be able to be climbed by a third party; and
 - o 3.0 metre clearance shall be maintained between the overhead line conductor and that part of the tree under/adjacent to the line that is able to support a ladder or be able to be climbed by a third party.
- For overhead lines at 66kV:
 - o 1.0 metre clearance shall be maintained between the overhead line conductor and that part of the tree under/adjacent to the line that is unable to support a ladder or be able to be climbed by a third party; and
 - o 3.2 metre clearance shall be maintained between the overhead line conductor and that part of the tree under/adjacent to the line that is able to support a ladder or be able to be climbed by a third party.
- For overhead lines at 132kV:
 - o 1.4 metre clearance shall be maintained between the overhead line conductor and that part of the tree under/adjacent to the line that is unable to support a ladder or be able to be climbed by a third party; and
 - o 3.6 metre clearance shall be maintained between the overhead line conductor and that part of the tree under/adjacent to the line that is able to support a ladder or be able to be climbed by a third party.

Clearances to effectively insulated¹ LV overhead lines are to be 1 metre from boughs that are climbable and 0.3 metre from other vegetation so as to prevent abrasion of the insulation.

¹ Effectively insulated overhead line shall be considered to be aerial bundled conductor (ABC)

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The return interval for vegetation cutting can be flexible depending upon the type of delivery method chosen. However the return interval needs to ensure that the minimum safe clearance is maintained between the overhead line conductor and the vegetation taking into account growth rates, species and the environment.

For situations where it has been identified that vegetation has encroached into these minimum clearances for non-effectively insulated overhead lines², the vegetation shall be removed within the following timescales:

Vegetation Response Matrix				
Encroachment type Site Status	Climbable Limbs within minimum safe clearance***	Vegetation growing through conductors (other than climbable limbs within minimum safe clearance)		Vegetation within minimum safe clearance to the side and/or under conductors
		LV	HV and above	
High Risk Site**	2 Weeks Maximum*	1 Month	1 Month	4 Months
Other Sites	1 Month	3 Months	1 Month	5 Months

* The above table is designed to indicate the maximum permissible timescales before encroaching vegetation shall be removed from overhead lines. Each incidence of vegetation encroachment shall be assessed on an individual basis to ascertain the level of risk posed. If the level of risk posed is deemed to be so great then the vegetation shall be removed immediately. Consideration shall however be given to the ownership of the vegetation in question as part of the decision making process.

** The definition of a "high risk site" is referenced in the Risk Assessment of Overhead Lines - DSS/023. The document provides guidance on the process employed for identifying sites where an overhead line route crosses the area or is in the vicinity. The assessment gives consideration to the use of the site (e.g. playing fields, picnic areas) and any evidence of activity on the site (e.g. vandalism, graffiti).

*** Minimum safe clearance is defined as the minimum clearance distance specified in NSP/004/011.

3.2.3 Opportunistic Resilience Clearance

In addition to the fundamental requirements outlined above, the opportunity shall be taken at each visit to undertake a deeper 'resilience clearance'. This is consistent with our obligation under the 2006 Regulations to avoid interference with and interruption to supply so far as is reasonably practicable. Specifically, where the extra work would increase the cost of each scheme by no more than 10%, the basic clearance required above shall be extended at the same visit to:

- Deliver a U rather than O cut; and
- Remove the risk from 'high risk' trees likely to come within falling distance of the line in the return interval, e.g. those that are diseased, damaged, or in poor ground.

² BLX and bare conductors shall be considered to be non-effectively insulated overhead lines

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3.2.4 Assessment of Resilience Clearances in Line with ENA-ETR 132

The approach to opportunistic resilience clearance outlined in the preceding section will reduce the overall risk to supply interruption from trees. However, it is inherently a span by span approach that may leave lines at risk where several spans pose low risk individually but a greater level collectively. To address this, a planned feeder by feeder management of resilience cuts, consistent with ENA-ETR 132³ is required. To achieve this:

- Asset Management (System Design) shall maintain a prioritised list of critical protection zones in accordance with ENA-ETR132 which shall include a consideration of but not limited to;
 - o The volume of customers potentially affected;
 - o The voltage level;
 - o The level of interconnection and options for providing alternate feeding arrangements;
 - o The level of comparable exposure between double circuit lines on single structures and dual circuits following separate geographical routes;
 - o The application of suitable weightings to provide a balanced view of exposure across all network configurations; and
 - o A cost estimate of securing falling clearance from 'high risk' trees for those lines.
- Asset Management (System Design) shall maintain a prioritised list of clearance activity, balancing the costs and benefits for each line. Strategies for vegetation removal shall include consideration of but not limited to:
 - o Where both feeds into a major substation or other system strategic position on the network are rated as high risk, removal of the vegetation which could impact network resilience only takes place on the circuit which provides the greatest cost/benefit. We shall however ensure that we secure NSP/004/011 clearance on both identified circuits at all times.
- Asset Management (System Design) shall provide this prioritised list as part of the annual investment planning process as per section 3.2.6.

3.2.5 Consideration of Other Mitigation Measures

For both basic and resilience clearance, an holistic approach shall be adopted, considering the options of:

- Cut and manage. The choice of how far to cut shall be based upon the economic balance between securing large clearances and long return periods against smaller clearance and shorter return periods. Once clearances have been established, it is expected that all future managed cuts will be carried out without the requirement for planned outages. Where insufficient clearance can be obtained to allow 'live' working on future managed cuts, consideration must be given to the effects that any planned interruptions have on both customers and any performance incentive scheme operating at that time.;

³ ENA-ETR 132 'Improving Network Performance Under Abnormal Weather Conditions by Use of a Risk Based Approach to Vegetation Management Near Electric Overhead Lines' is designed to present a risk based approach to provide proactive and positive impact on network performance by improving the resilience of the network suffered by vegetation related faults.

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- Clear fell where economic for the company and acceptable to the landowner. This may be justified on the basis either of the costs of return visits or upon an assessment of 'high risk' trees in accordance with ENA-ETR 132;
- Rebuild with covered conductor. Particularly at LV, there are benefits from reduced ongoing need for tree trimming, planned interruptions and synergies with other work programmes. Credit shall be given for any benefit of improved asset condition, e.g. replacement of corroded conductors;
- divert, on alternate overhead or underground routes. The same issues apply as for covered conductor, although the costs will be higher. Introducing underground sections into predominantly overhead circuits, particularly at HV, increases the risk of permanent damage from lightning and potentially greatly increases restoration time. This may therefore be considered as a last option; or
- Erect a recloser where an assessment of resilience clearance identifies that vegetation on a spur line threatens security of a main line.

In general, the greater the costs of clearing trees, the greater the difficulty of obtaining adequate clearance to allow for a reasonable return frequency and the greater the requirement for planned interruptions, then the more economic it will be to invest in the line to remove or reduce the problem. All schemes costing more than £5 per metre for LV and £18 per metre for HV and above to cut vegetation to achieve the desired clearance shall have alternate solutions assessed.

3.2.6 Review of Investment Levels

Asset Management shall carry out an assessment of the level of risk posed by vegetation to the distribution network as part of the annual asset serviceability review. In the event that the review demonstrates the level of risk has significantly increased due to a reduction of clearances of vegetation from overhead lines, then the report shall inform a decision on the required levels of capital investment. This shall ensure that the desired level of risk due to vegetation clearances from overhead lines is achieved and maintained.

Specifically, the activity levels enhanced above NSP/004/011, as required for compliance with ENA-ETR 132, shall be monitored and compared to their impact on network resilience.

3.2.7 Work Delivery

The application of vegetation management in the vicinity of our overhead lines can, if not properly administered, have a negative impact upon the environment and the customers who we serve. This can have a direct influence on our company goal of Environmental RESPECT, and restrict our ability to erect and manage overhead lines in the future. In order to avoid such incidents all vegetation related works carried by or on behalf of CE UK shall be in compliance with ENA-ETR 136 'Vegetation Management Near Electricity Equipment – Principles of Good Practice'.

The purpose of this document is to provide generic guidance to practitioners when carrying out works in pursuit of our legal obligations in order to generate good customer co-operation and an understanding as to why vegetation management is necessary.

Wherever possible, cutting will be carried out without interrupting the supply to customers. Where interruptions are unavoidable, the opportunity to co-ordinate with other outages on the overhead line must be taken. It is not acceptable to subject customers to repeated planned interruptions over short timescales.

3.2.8 Data Requirements

There are a number of sources that require information with regard to vegetation that has the potential to or is affecting the overhead network. To manage these requirements the following data shall be made available for all overhead networks as part of service delivery:

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- Pole number, feeder number and line voltage;
- The geographical location of the vegetation in respect of spans affected;
- Date inspected, date cut and next return visit;
- Spans where vegetation does not pose any risk;
- The number of spans with the potential to be affected by vegetation;
- Cutting information to be provided by span, circuit and length;
- Vegetation Species – Type, growth rate, environment;
- Restricted cuts - E.g. Tree owners' information, preservation orders and visual amenity;
- Clearance from line taking account of NSP/004/011 where full clearance cannot be achieved due to other influences; and
- 'High Risk' areas where vegetation encroachment into overhead lines could cause risk to public safety;
- Areas where cutting with the overhead line 'live' would be unsafe.

In terms of resilience clearance work the following additional information is required:

- Potential tree falling distances;
- Diseased trees;
- Damaged trees; and
- Unstable ground.

3.3 Assumptions

The following assumptions have been made but will be required to ensure the effective delivery of this policy into the business:

- The current statutory clearances from vegetation remain unchanged.
- The current business goals for the security, safety and continuity of operation of the network do not change.

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3.4 Monitoring and Control of this Policy

The roles and responsibilities for the implementation and monitoring of this policy are detailed in the table below:

Designation	Role
Construction Manager	Company wide responsibility for the management and control of the vegetation management process. Company wide responsibility for identifying opportunities to improve the current process and procedure for vegetation management.
Design Manager	Assessment of incremental investment in the vegetation management activity.
Safety Manager	Monitor compliance of those accountable for implementation of the vegetation management policy.

3.5 Planned Policy Review

This policy shall be proposed for review on a biennial basis or at any time when external or internal influences drive a change in policy e.g. a change in legislation. Additionally, following the update of other relevant documentation, this policy will be reviewed to ensure the appropriate cross-reference is maintained.

The following responsibilities shall apply to policy control and review:

Designation	Responsibility
Publication Manager	Responsible for issuing a quarterly report to the Policy Production Manager (or representative) detailing policies scheduled for biennial review within the next 6 months.
Policy Production Manager	Responsible for assessing the continued applicability of this company policy and for amending this document and communicating any changes in policy.

3.6 Superseded Documentation

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Title	Version
DSS/024	Policy for the Management and Control of Vegetation Near Overhead Lines	V1.0 Nov 2003

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4 References

4.1 External Documentation

Reference	Title	Version and date
ESR 88 regs	<i>The Electricity Supply Regulations 1998 No 1057</i>	Oct 1988
ESQCR 02 regs	<i>The Electricity Safety, Quality, and Continuity Regulations 2002 No. 2665</i>	Jan 2003
ESQCAR	<i>The Electricity Safety, Quality, and Continuity (Amendment) Regulations 2006 No. 1521</i>	Jun 2006
	<i>Health and Safety at Work Act 1974</i>	1974
06 regs	<i>The Electricity Safety, Quality, and Continuity Regulations 2002 No. 2665 as amended by the Electricity Safety, Quality, and Continuity (Amendment) Regulations 2006 No. 1521</i>	
	<i>Electricity at Work Regulations 1989 No 635</i>	1989
ENA-TS 43-8	<i>ENA Technical Specification 43-8 Overhead Line Clearances</i>	Issue 3, 2004
ENA-ETR 132	<i>Engineering Technical Recommendation 132 Improving Network Performance under Abnormal Weather Conditions by Use of a Risk-Based Approach to Vegetation Management Near Electric Overhead Lines</i>	Mar 2006
ENA-ETR 136	<i>Engineering Technical Recommendation 136 Vegetation Management Near Electricity Equipment – Principles of Good Practice</i>	Jan 2007

4.2 Internal documentation

Reference	Title	Version and date
NSP/004/0 11	<i>Guidance on Overhead Line Clearances</i>	V1 Jul 2006
DSS/023	<i>Risk Assessment of Overhead Lines</i>	V1 Oct 2003
MNT/004	<i>Inspection and Maintenance of Overhead Lines</i>	V1 Nov 2003

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5 Definitions

Term	Definition
YEDL	<i>Yorkshire Electricity Distribution PLC</i>
NEDL	<i>Northern Electric Distribution Limited</i>
Ofgem	<i>Office of gas and electricity markets</i>
ABC	<i>Aerial Bundled Conductor</i>
QoS	<i>Quality of Service, i.e. customer interruptions</i>
BERR	<i>Department for Business, Enterprise and Regulatory Reform</i>
Effectively Insulated Conductors	<i>A line conductor which is insulated for continuous phase to phase or phase to earth contact and is protected, so far as is reasonably practicable, against mechanical damage or interference having regard to its accessibility. Note: The implication here is that insulated conductors (such as low voltage aerial bundled conductors but not high voltage CC conductors) may be placed such that they are ordinarily accessible, providing they are safe in the particular circumstances.</i>

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6 Authority for issue

1. Policy Author

I sign to confirm that I have completed and checked this policy and I am satisfied with its content

		Sign	Date
Paul Hollowood	Asset Management Engineer	Paul Hollowood	16/06/08

2. Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this policy and submit it for approval and authorisation.

		Sign	Date
Andrew Ellam	Network Performance and Risk Manager	Andrew Ellam	16/06/08

3. Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this policy and submit it for approval and authorisation.

		Sign	Date
Jim Morrell	Senior Asset Management Engineer	Jim Morrell	19/06/08
Dave Miller	Senior Asset Management Engineer	Dave Miller	20/06/08
Sean Caskey	Network Construction Manager	Sean Caskey	04/08/08

4. DBD Assurance

I sign to confirm that this policy has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	Safety Engineer	Sean Johnson	08/08/08

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5. Approval

Approval is given for the content of this policy

		Sign	Date
Mark Nicholson	Head of System Strategy	Mark Nicholson	16/06/08
Peter McCormick	Director of Group Health, Safety and Environment	Peter McCormick	13/08/08

6. Authorisation

Authorisation is granted for publication of this policy

		Sign	Date
Mark Drye	Director of Asset Management	Mark Drye	29/08/08